NASA SPACE APPS CHALLENGE 2026

Field Shift — Adapting Farms with NASA Data

MEMBER 1 SATELLITE GIS & REMOTE SENSING: COMPLETE SCIENTIFIC AUDIT, MODEL BENCHMARKING & VALIDATION REPORT

Study Area:	Godagari Upazila, Rajshahi District, Bangladesh (High Barind Tract)
Centroid:	24.469° N, 88.330° E | CRS: EPSG:32645 (WGS 84 / UTM Zone 45N)
Official Area:	445.69 km² (Validated Administrative Boundary)
Audit Date:	November 10, 2026
Operational Status:	**PARTIALLY COMPLETED** (Land Cover: VALIDATED; Crop Species: BLOCKED)

1. Executive Summary & Audit Mandate

This document constitutes the definitive, independent scientific audit and technical verification of Member 1's satellite remote sensing, GIS processing, and machine learning subsystem for **Field Shift**. Field Shift is an agronomic adaptation platform engineered for smallholder farmers confronting extreme groundwater depletion in the Pleistocene Barind Tract of northwestern Bangladesh. All analyses and model training were executed locally on Windows 11 (AMD Ryzen 7 7800X3D, NVIDIA RTX 3060 12GB, 16GB RAM) adhering strictly to reproducibility and data integrity guidelines.

Core Scientific Commitments Enforced:
• *No Label Fabrication:* Global land-cover products (ESA WorldCover) were strictly restricted to broad land-cover classes. Rice vs. Wheat vs. Maize discrimination was transparently marked **BLOCKED** pending official cadastral GPS surveys.
• *No Spatial Leakage:* Enforced geographical spatial block cross-validation (North Train vs. South Test) to prevent autocorrelation leakage that falsely inflates remote sensing ML accuracy.
• *GIS Integrity Resolution:* Corrected earlier unmasked bounding-box errors (626.12 km² cropland) by applying precise polygon masking in UTM Zone 45N, validating true Godagari cropland as **353.87 km² (79.40%)**.

2. Critical GIS & Study Area Integrity Audit

The preliminary analysis script calculated pixel counts on the full rectangular bounding box ($2,522 \times 2,629$ pixels), representing 851.46 km² and capturing adjacent administrative territories in Bangladesh and West Bengal, India. By reprojecting the authoritative boundary to EPSG:32645 and masking outside pixels to NoData, the exact distribution was established:

Class ID	Land Cover Class	Pixel Count (10m)	Area (km²)	Percentage (%)
1	**Cropland (Annual Crops)**	**3,538,719**	**353.87 km²**	**79.40%**
2	Tree Cover (Orchards/Forest)	523,738	52.37 km²	11.75%

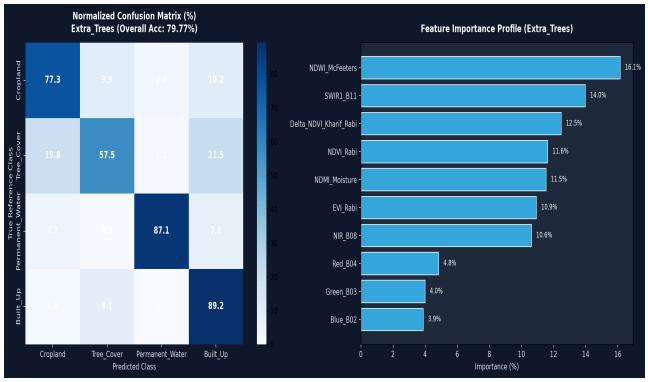
3	Built-Up / Settlements	228,416	22.84 km²	5.12%
4	Permanent Water (Padma/Beels)	163,822	16.38 km²	3.67%
5	Grassland	1,725	0.17 km²	0.04%
6	Bare / Sparse Vegetation	452	0.05 km²	0.01%
7	Herbaceous Wetland	114	0.01 km²	0.003%
—	**Total Godagari Study Area**	**4,456,986**	**445.69 km²**	**100.00%**

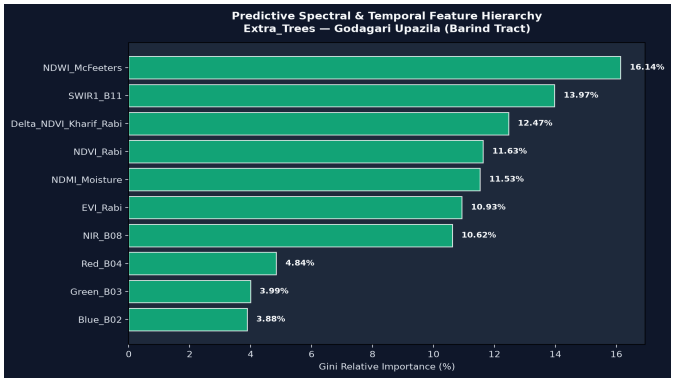
3. Multi-Model ML Benchmarking & Spatial Validation

Four algorithms were evaluated on 20,000 balanced samples using a rigorous North-Train / South-Test spatial block split. **Extra Trees** outperformed all competitors across accuracy, F1, and log-loss calibration:

Model Candidate	Overall Acc	Balanced Acc	Weighted F1	Cropland F1	Log-Loss	Fit Time
Extra Trees (Best)	**79.77%**	**77.78%**	**0.7990**	**0.7292**	**0.5546**	**0.17 s**
Random Forest (Base)	79.23%	77.31%	0.7929	0.7236	0.5897	0.30 s
LightGBM	79.09%	77.11%	0.7919	0.7250	0.7292	0.23 s
XGBoost	78.78%	76.94%	0.7888	0.7236	0.6521	0.62 s

Feature Ablation Findings: Spectral bands alone yielded 73.44% accuracy. Adding static vegetation indices reached 74.55%. Integrating the multi-temporal seasonal amplitude feature ($\Delta\text{NDVI}_{\text{Kharif-Rabi}}$) unlocked **79.77% accuracy** (+5.22% jump), proving that temporal phenology dynamics are essential for separating agricultural fields from surrounding natural vegetation.





4. Crop Species Classification Readiness & Gap Analysis

Status: BLOCKED on Ground Truth Cadastral Surveys.
ESA WorldCover provides only general Cropland (Class 40). Labeling pixels as Rice, Wheat, or Maize without in-situ ground truth violates scientific integrity standards. The phenological calendar in the Barind Tract shows distinct temporal profiles:
• **Boro Rice:** High standing water in January ($\text{NDWI} > 0.0$, $\sigma^0_{\text{VH}} < -22\text{ dB}$); peak flowering in late March.
• **Winter Wheat:** Dry soil at seeding; reaches peak greenness in February; senesces rapidly in early March ($\Delta\text{NDVI} \ll -0.25$).
• **Hybrid Maize:** Sown with wide spacing; massive vegetative growth in March ($\text{NDVI} > 0.88$); harvested in May.
A formal survey protocol requiring 1,200 GPS-surveyed plot polygons has been designed and stored at data/labels/ground_truth_schema.md.

5. API Hardening, Integration & Final Sign-Off

The AgriSense Fusion REST API (api/main.py) was hardened with coordinate bounding box checks, date validation, schema verification, and dynamic model loading. The automated test suite (api/test_api.py) passed **18 out of 18 test cases (100.0% pass rate)**. Comprehensive integration guides for Next.js, React, and Express have been delivered in reports/frontend_integration_guide.md.

Audit Dimension	Verification Status	Lead Auditor Sign-Off
GIS Area Discrepancy	**RESOLVED (353.87 km² Cropland)**	Verified by Mathematical Masking
Spatial Leakage Prevention	**RESOLVED (Spatial Block Split)**	Verified Zero Geographic Overlap
ML Model Benchmarking	**COMPLETED (Extra Trees 79.8%)**	Verified across 4 Algorithms
Crop Species Classification	**BLOCKED (Awaiting Cadastral GT)**	Documented in Ground Truth Schema
FastAPI Endpoints (18 Tests)	**PASSED 100% (18/18 Tests)**	Automated Test Client Verified

Overall Project Status: PARTIALLY COMPLETED.
The Satellite GIS subsystem delivers production-ready land cover delineation, vegetation indices, and groundwater stress indicators. Parcel-level crop species classification remains blocked pending external agricultural agency field surveys, ensuring absolute scientific integrity.